

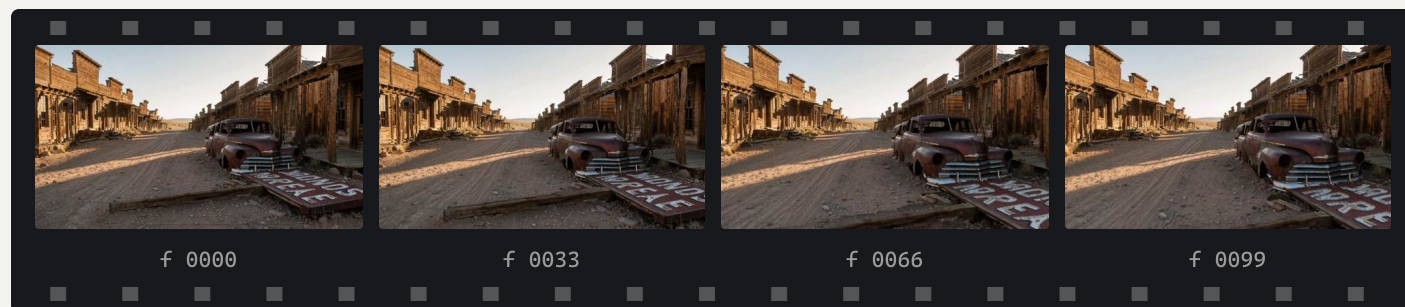
ELEVEN WORKFLOWS, EVERY NODE, RUN-VERIFIED

PLATES 9 WORKFLOWS 11 NODE COVERAGE 60 / 60 VERIFIED 2026-07-16 ENGINE ComfyUI

The operator's field guide to the shipping showcase set: what each workflow demonstrates, the exact result it produced on a real run, and — most importantly — **which knobs you retune when you swap in your own plate**. Generated by `tools/generate_showcase_workflows.py`; edit the generator, never the JSON.

READ THIS FIRST: THE GATES ARE NOT A BUG

Every workflow ships with its checkpoints **closed**. The first Queue runs the cheap solve preview in seconds, then stops. Review the solve, click ☒ **Approve Solve** on the gate node (and **Continue Workflow** on the VLM node where present) — the workflow re-queues itself and the heavy stack runs. Hitting Queue again without the button just repeats the preview.



One still photo → a real camera move. → **Dolly In**, 100 frames @ 24 fps at 1920×1080 — instant-relief mode (**layers=0**, ultra mesh), the clean take for a continuously-receding street. See workflow 08.



coastal_alley_for_vlm.png · 1408×768

The teaching workflow. A geometric vanishing-point solve with every diagnostic laid out: VP fan-line overlay, horizon and ground masks, analytic ground depth, the decompose nodes, a solve-JSON save→reload round-trip, and the full three-tier metric-scale ladder — known-size door, VLM cues, assumed default. A "legacy corner" shows the deprecated path-based solve so you know what not to build on.

AtlasSolveFromImage

AtlasVPVisualization

AtlasReferenceScaleSolve

AtlasVLMscaleCues

AtlasConstrainedSolve

AtlasGroundDepthMap

RETUNE ON YOUR PLATE

- ▶ The door's `bbox_x0/y0/x1/y1` on `AtlasReferenceScaleSolve` are pixel coordinates drawn for *this* image — redraw them, and pick the right `reference_id` (door / person / car).
- ▶ `AtlasConstrainedSolve`'s line-groups JSON traces this alley's edges; re-author for a new plate or bypass it.
- ▶ The VP solve needs strong straight lines — on organic or AI plates, switch to the learned solve.
- ▶ VLM provider/model on `AtlasVLMscaleCues` if you're not on lmstudio :1234.

RUN ✓ VP solve **85° hFOV**; door reference set camera height to **1.02 m**; VLM cues returned zero references and the confirm-gated apply correctly adopted nothing.

[Deep dive →](#)

02 CITY BLOCKS — AERIAL PRESET



newyork_Birdseye.png · real photo · 7360×4912

A real 177 MB aerial photo through the one-node `scene_type=aerial` preset — building boxes over a relief ground — conformed to a delivery format with ShotCam and handed to Maya as a review scene with the real relief OBJ, plus the full review package.

AtlasDeriveProjectionGeometry

AtlasScaleOverride

AtlasDefineShotCam

AtlasExportMayaReviewScene

AtlasExportReviewPackage

RETUNE ON YOUR PLATE

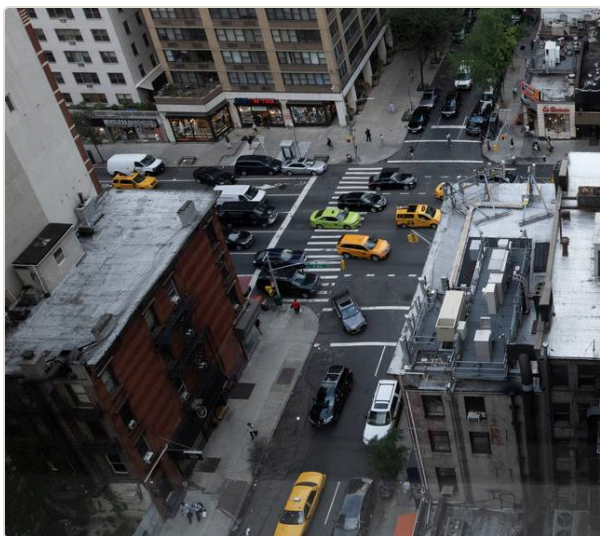
- **Count the storeys — the sky-rise scale doctrine.** Elevated vantages break the ground fit and the solve falls back to 1.6 m. The workflow ships `AtlasReferenceScaleSolve` on a counted tenement ($5 \times 3.5 \text{ m} = 17.5 \text{ m}$) → camera locked at **63.7 m** by single-view geometry. The original eyeballed 25 m dial was $\sim 2.5\times$ small against the photographer's ground truth — count, don't guess.
- `AtlasDefineShotCam` — your delivery sensor / focal / resolution.
- `max_objects` on the derive node if the box count reads sparse or busy.

RUN ✓ 7 fitted primitives + relief at real-world metres; viewport conformed to 1920×1280; both packages written. Browser preview of the plate is heavy — the server side is fine.

[Deep dive →](#)

03 THE CLASSIC DMP ANGLE, PART 1 — ANCHORED WALLS

atlas_dmp_angle_anchored_newyork.json



same plate · the street-contact problem

Foreground buildings meet the street in-frame and ground naturally — but background bases are hidden behind rooflines, so the relief tears and those buildings *float*. The fix: wall derivation with `ground_anchor=True`, which trusts the exact ray geometry where a facade visibly meets the ground and demotes the depth model to grouping pixels, merged back over the relief.

AtlasDeriveWalls

AtlasDepthMap

AtlasMergeGeometry

AtlasScaleOverride

RETUNE ON YOUR PLATE

- `distance_modes = 3` — more modes = more depth rows per facade direction; raise on deep street grids.
- `max_walls` for dense skylines; keep the scale override at your estimated height.
- Fully-hidden bases are *refused* by the anchor's safety gates — that residual is deliberate, and it's exactly what Part 2 exists for.

RUN ✓ 13 anchored facades, every one planted at $Y \approx 0$, rising 22–54 m = 6–15 storeys — countable against the photo, independently validating the anchor geometry.

[Deep dive →](#)

04 THE CLASSIC DMP ANGLE, PART 2 — X-RAY

EXPERIMENTAL

atlas_dmp_angle_xray_newyork.json



same plate · what's behind the rooflines?

LaRI predicts the occluded lower floors behind the near rooflines. The key discipline: the prediction is **restrict-masked to the foreground depth band** ([0–45%] → `restrict_mask`), so it only substitutes geometry where the occluder actually is. LaMa supplies the reveal pixels for a mask-membership X-ray layer.

AtlasPredictHiddenGeometry

AtlasDepthLayerMask

AtlasCleanPlateLayer

LaMa inpaint

RETUNE ON YOUR PLATE

- The depth band's `far_pct` (0.45) — widen or narrow to cover exactly your occluding foreground.
- Server must launch with
`ATLAS_EXPERIMENTAL=1` +
`ATLAS_LARI_PATH` pointing at a LaRI clone.
- Architecture is LaRI's strong domain; expect weaker predictions on open terrain (see workflow 11).

RUN ✓ Registration rel-MAD **0.113** — the best of any plate; X-ray layer 56k verts with matte + provenance; substitution confined to the fg band. [Deep dive →](#)

03 — GEOMETRY COMPOSITING

05 COMPOSABLE GEOMETRY — THE MERGE CHAIN

atlas_composable_geometry_templecity.json



atlas_00022_templecity.png · 9216×3840

Nuke-style geometry compositing: one shared depth map fans into **Walls** (3 distance modes) + **TowersSpires** (roofline split reaches the spires) + **RoofsFacades** (RANSAC planes) + relief, combined through a chain of three merge nodes into a single-projection Nuke scene with the real relief geometry.

AtlasDepthMap

AtlasDeriveWalls

AtlasDeriveTowersSpires

AtlasDeriveRoofsFacades

AtlasMergeGeometry ×3

AtlasExportNuke

RETUNE ON YOUR PLATE

- `AtlasScaleOverride camera_height_m = 16` — same per-plate scale dial as NYC.

- Per-branch: `distance_modes` on Walls, `roofline_split` on TowersSpires, `max_planes` on RoofsFacades. Bypass branches you don't need — the merge chain combines whatever survives.
- Merge order matters: `solve_a`'s camera wins — keep the relief branch first.

RUN ✓ 18 merged primitives; the .nk carries Camera2/Project3D2/ReadGeo2 + the OBJ.

Known cosmetic: 3-way merges keep duplicate ground planes (identical texels, harmless).

[Deep dive →](#)

06 OCIO LAYERED DMP — OCEAN CASTLE

`atlas_ocio_dmp_oceancastle.json`



`oceancastle_32bit_acescg.exr`

The professional handoff: EXR read in ACEScg, plate registered so exporters point at the float original, VLM assess → solve gate, SAM3 sky → sky dome, and **one absolute band split driving both clean-plate layers** — fg and bg resolve the identical metric boundary by construction, so the bands can never drift apart. Per-layer Nuke scene + USD out, all referencing the original EXR.

`OCIORead`

`AtlasRegisterPlate`

`AtlasAttachSourcePlate`

`AtlasDepthBandSplit`

`AtlasSkyDomeLayer`

`AtlasExportNukeLayers`

RETUNE ON YOUR PLATE

- `OCIORead.source` and `AtlasRegisterPlate.plate_path` are **absolute, per-machine paths** — repoint both at your EXR.
- The band split's percentile — where your foreground subject ends and background

begins.

- SAM3 sky prompt if your plate's sky isn't a plain "sky".
- Layer priorities stay farthest-highest (sky -10 / castle 0 / sea 10) — don't invert them; that ordering is what makes the seams render correctly.

RUN ✓ 3 layers exactly as designed: sky 1.6k / castle 54k / sea 80k verts; per-layer plates, mattes, OBs + .nk + camera.usda written.

[Deep dive →](#)

07 INTERIOR DOCTRINE — SPACE HANGAR

atlas_interior_hangar.json



spacehangar_32bit_acescg.exr

Everything interiors need: V2-Indoor depth, MoGe relight normals decoupled from the depth source, a Manhattan room fit merged with relief, a semantic-mask "sky" self-check (~0% on an interior — proving the mask), and the -gated render-conditioned patch loop. Hands off to Maya as per-layer projection networks.

AtlasDeriveInteriorRoom

AtlasMogeNormals

AtlasSemanticMask

AtlasOcclusionMask

AtlasAddPatchView

AtlasExportMayaLayers

RETUNE ON YOUR PLATE

- ▶ **Two gates:** the VLM's ▶ Continue (automatic here), then  Approve Solve.
- ▶ Depth model: swap **V2-Indoor** ↔ MoGe to taste — MoGe is often cleaner on enclosed shots.
- ▶ The patch branch stays **paused by design** until you orbit the viewport and click  Extract Angle — that's the intended interactive step, not a hang.
- ▶ **AtlasMogeNormals normal_model:** **vitl** best, **vits** for CPU.

RUN ✓ Relight layer carries the MoGe **normal map** (verified in the payload); patch branch correctly paused; .ma + layer assets written.

[Deep dive →](#)

08 ONE NODE → A CAMERA MOVE — GHOST TOWN

atlas_input_cameramove_ghosttown.json



ghosttown_32bit_acescg.exr

The hero pipeline (it made the filmstrip up top): the all-in-one `AtlasInput` node — VLM plan, sky, 4 depth bands, LaMa clean plates — into a detached Output Desk, a one-click camera move, ● Bake, and out come a JPEG sequence, an MP4, and a time-sampled USD camera. A debug report writes structured JSON every run so nothing fails silently.

`AtlasInput`

`AtlasViewportControls`

`AtlasDebugReport`

`AtlasExportCameraPathUSD`

`VideoCombinePlus`

RETUNE ON YOUR PLATE

- ▶ `AtlasInput` : `layers=4`, `sky=True` are set **explicitly** (don't rely on `use_vlm` to force them); `vlm_scope` ships **OFF** — ON can gut a band on a partial segment match; `edge_extend_px=24` .
- ▶ **Know when to skip the bands entirely.** Depth-banding assumes separable layers; a continuously-receding street isn't one — the VLM's flat far card slices the midground, and inpaint fill smears through the seams. For scenes like this, `layers=0` (one full-depth mesh, `mesh_resolution=1024` for ultra) is the clean take — it's what the filmstrip at the top was baked with.
- ▶ Move choice: → **Dolly In is the clean take** (on-axis, minimal disocclusion). Orbit shows real parallax but exposes inpaint smear on reveals — ship it as the honest comparison.
- ▶ 📡 lens slider: left = wide enough to see the geometry's edges; playback-only, never touches the solve.
- ▶ The real frame sequence comes out of the bake — the in-graph saver writes one preview file per queue, by design.

RUN ✓ Full 5-layer master; baked **100 frames @ 24 fps, 1280×720** + MP4 + camera_path.usda.
This workflow found and fixed two real bugs during verification.

[Deep dive →](#)

09 ORGANIC RELIEF + RETOPO — JUNGLE RUINS

atlas_organic_relief_jungleruins.json



jungleruins_32bit_acescg.exr

Canopy doctrine: relief at grid 512 with a loosened edge threshold (1.0) so dense foliage doesn't shred into holes, then export-only hole-fill and **quad retopology** down to ~4.5k verts — a mesh you can actually sculpt on — out as textured OBJ + GLB plus a Blender build script.

AtlasDeriveReliefMesh

AtlasBoundedBand

AtlasExportReliefMesh

AtlasExportBlender

RETUNE ON YOUR PLATE

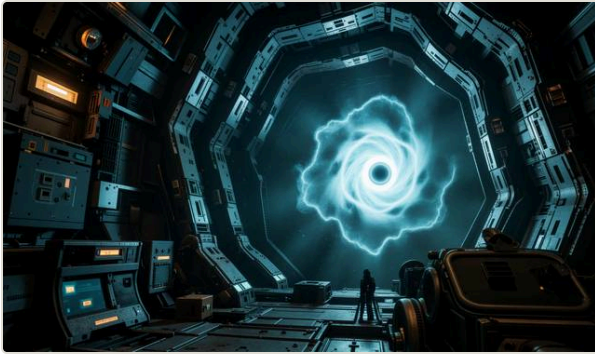
- **relief_grid** / **depth_edge_rel** — the organic dial pair: denser canopy wants a looser edge threshold.
- **retopo_method** (quad / decimate / smooth) and **retopo_target_vertex_count** on the export node.
- **fill_interior_holes** + the **fill_depth_near/far_m** band box to scope which tears get capped.
- Wire the export's **preview_solve** into a viewport to eyeball the fill before a DCC round-trip.

RUN ✓ Quad retopo landed at **4,493 verts** (target 4,000); OBJ/GLB import textured — projective UVs regenerated; build_scene.py ready for Blender.

[Deep dive →](#)

10 ROLL TRIM + ANAMORPHIC CONFORM — PORTAL

atlas_rolltrim_shotcam_portal.json



atlas_00024_portal.png · 7680×4512

The fix for "the grid doesn't match the interior lines": the roll-trim dial levels a solve whose learned gravity drifted a few degrees (-3° was the measured correction on this plate — the classical VP detector found nothing to catch it with). Then MoGe relief, a 2.39:1 anamorphic ShotCam conform, and a USD camera round-trip that proves the export decomposes back to the same lens.

AtlasRollTrim

AtlasDefineShotCam

AtlasExportUSD

AtlasUSDCameraLoader

AtlasDecomposeCamera

RETUNE ON YOUR PLATE

- ▶ **AtlasRollTrim roll_deg** — **entirely per-plate, tuned by eye** against architecture verticals. Positive turns the scene counter-clockwise on screen.
- ▶ It must sit *between* the solve and the derive nodes — geometry back-projects through the corrected view.
- ▶ ShotCam sensor / focal / resolution for your delivery format.

RUN ✓ Viewport conformed to 2048×856 — exactly 2.39:1; the round-tripped USD camera decomposes to the same lens.

[Deep dive →](#)

11 X-RAY WITH A RESTRICT MASK — THE WRECK

EXPERIMENTAL

atlas_xray_wreck.json



atlas_00025.png · 7680×4512

What's behind the rusty machine? LaRI predicts it — but unconstrained, it wanted to substitute half the frame including visible background. A SAM3 segment of the occluder as `restrict_mask` is the whole trick: substitution drops to just the machine. LaMa supplies the reveal pixels; the render-repair node ships bypassed (it needs a Docker environment).

AtlasPredictHiddenGeometry

SAM3Segment

AtlasCleanPlateLayer

AtlasRenderFix (bypassed)

RETUNE ON YOUR PLATE

- ▶ The SAM3 prompt ("rusty machine") — describe *your* occluder. The restrict mask is not optional in practice.
- ▶ Open-terrain backgrounds are LaRI's documented weak domain — this plate ships specifically to show that honestly.
- ▶ Un-bypass `AtlasRenderFix` only if you've built the Fixer Docker image.

RUN ✓ Restrict mask cut substitution from **50.6%** → **23.8%** of frame; registration rel-MAD 0.190 ("good").

[Deep dive →](#)

CROSS-CUTTING

THE KNOBS THAT MATTER ON EVERY PLATE

- **The scale fix.** Any elevated vantage or AI-generated vista solves to an assumed 1.6 m eye height (~10× small). Best: **count a building's storeys** (×3.5 m) into `AtlasReferenceScaleSolve` — tier-1 measured scale (NYC: counted tenement → camera 63.7 m). Fallback: the `AtlasScaleOverride` dial (temple city ≈16 m). Every downstream metre follows.
- **Gate buttons before anything heavy runs.** Queue → review → Approve Solve. A 2-second "successful" run means a gate is closed, not that the workflow is broken.
- **EXR paths are absolute and per-machine.** Repoint `OCI0Read.source` and `AtlasRegisterPlate.plate_path` together.

- **If a layer looks empty**, read `atlas_debug/master_debug.json` — the debug node names the offending layer, its vertex count, and any band gaps directly.
- **Depth doctrine**: exteriors → V2-Outdoor; interiors → MoGe or V2-Indoor. MoGe's far field runs away outdoors; pair V2 depth with `AtlasMogeNormals` when you want its relight normals anyway.

REQUIREMENTS AT A GLANCE

NEEDS	WHICH WORKFLOWS	NOTES
Atlas + [neural]	all	plates staged in ComfyUI/input/ (any photo of your own works)
OCIO / EXR	06 · 07 · 08 · 09	ComfyUI-OCIO, opencv-python 4.x, and <code>OPENCV_IO_ENABLE_OPENEXR=1</code> set before python starts
VLM (lmstudio)	01 · 06 · 07 · 08	vision model on :1234 (gemma-4-12b preset); everything fails soft without it
Experimental env	04 · 11	<code>ATLAS_EXPERIMENTAL=1</code> + <code>ATLAS_LARI_PATH</code> → LaRI clone (research-only license)
Third-party packs	varies	KJNodes, RMBG (SAM3), inpaint-nodes + big-lama.pt, pysssss, rgthree, VideoCombinePlus
[moge] / [usd] / pyinstantmeshes	07 · 09 · 10	MoGe depth & normals · USD exports · quad retopo

Atlas Camera v0.6.0 beta · every workflow above ran to completion on 2026-07-16 ·
examples/showcase/README.md carries the full findings log